Overfitting

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 24: validation guides model choice, final test evaluates it.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Compare training and validation evidence for model capacity. Try the recall before opening notes.

Your learning target

I can compare **training** and validation **evidence** for model **capacity**.

I can explain **overfitting** without assuming every gap proves it.

I can use a declared selection rule instead of perfect **training** fit.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

An overfit model captures **training**-specific noise, producing a large gap between **training** and unseen performance.



Your target

LEARNING TARGET A has **training** 98%, validation 58%; B has **training** 86%, validation 81%. Compute gaps and select a candidate with a qualification.

Builder name and experiment date:

Readiness for the connected explanation

Can a zero **training** error guarantee good validation performance?

This week's end goal

By the end, I can explain **overfitting** using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Overfitting case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **overfitting** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A more flexible model can represent complicated relationships and accidental details. **Capacity** can help when there is enough relevant **evidence**, but perfect **training** performance is not the goal.

Compare **training** and validation using the same metric while checking leakage and representativeness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **overfitting** and model **capacity**.

STUDENT LESSON · 3 OF 18

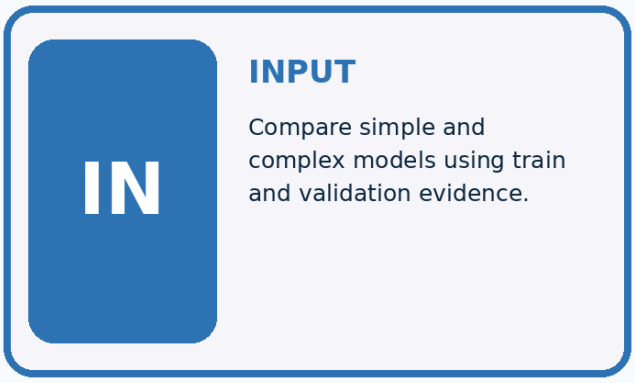
See how it works

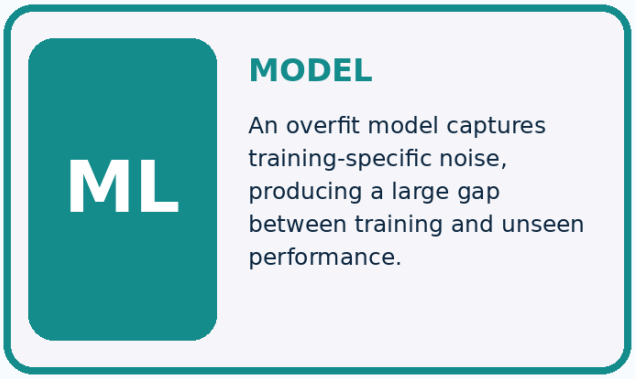
> Follow the mechanism for Overfitting. Connect each labelled input, deciding step and result.

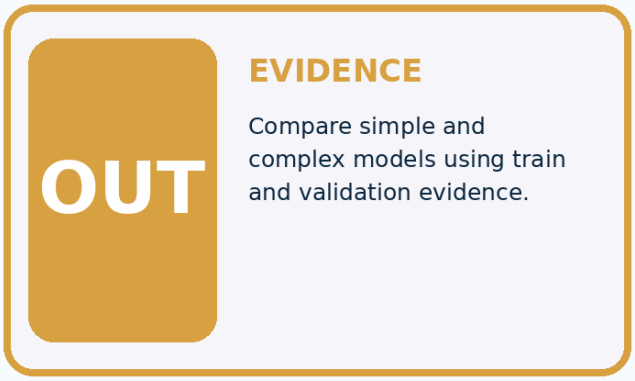
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Compare simple and complex models using train and validation **evidence**.

MODEL IDEA An overfit model captures **training**-specific noise, producing a large gap between **training** and unseen performance.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Simple tree: train 82%, validation 78%; gap 4 points. Deep tree: train 100%, validation 65%; gap 35 points.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

On the same regression split, a constant has **training** MAE 5 and validation 6, a line has **training** 2 and validation 3, and a highly flexible curve has **training** 0 and validation 8. Under smallest validation MAE, choose the line.

The curve has the lowest **training** loss but the worst validation result.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

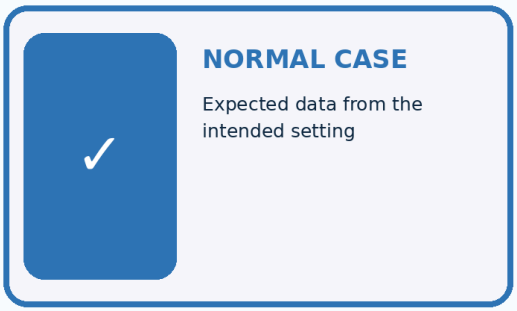
Compare the cases

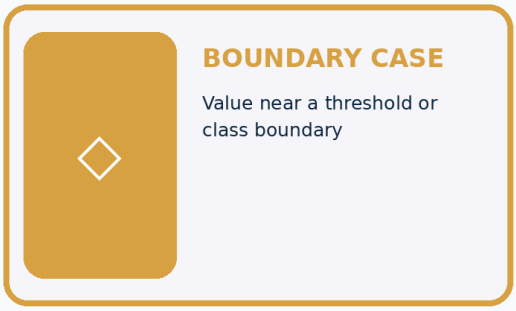
> Compare the supplied cases for Overfitting. Identify the detail that changes the conclusion.

EVALUATION DESIGN

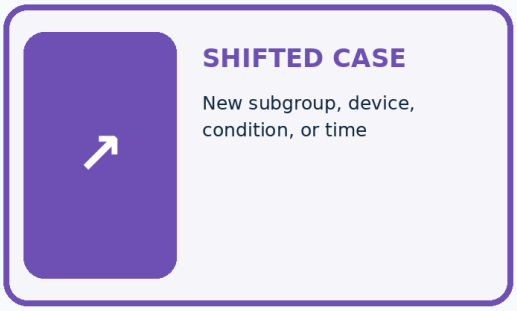
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves **generalization**. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Large's validation accuracy rises to82% after a development revision.

Predicted result

Prediction: Its gap is 18 percentage points; it now beats Small on this validation set.

Why the result follows

Reason: Validation performance improved, but the **training**-validation gap is not a complete release rule.

STUDENT LESSON · 5 OF 18

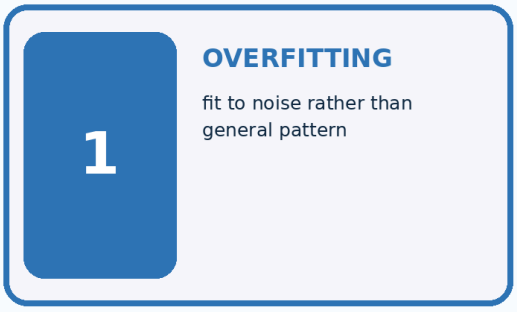
Words that unlock the idea

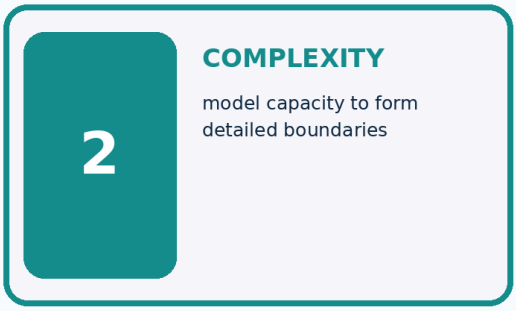
> Use the vocabulary of Overfitting to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
overfitting	fit to noise rather than general pattern	
complexity	model capacity to form detailed boundaries	
generalization gap	train score minus unseen score	
regularization	constraint discouraging excessive complexity	

MISCONCEPTION TO REPAIR The model with the highest **training** accuracy is the best model.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Capacity	Flexibility of a model to represent patterns
Overfitting	Learning details that fail to generalize well
Underfitting	Failing to capture useful structure
Generalization	Performance on relevant unseen examples

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Overfitting, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 27 • WORKED EVIDENCE

1. Simple tree: train 82%, validation 78%; gap 4 points.
2. Deep tree: train 100%, validation 65%; gap 35 points.
3. Deep tree's perfect training result is weaker unseen evidence.
4. Prune, regularize, improve representative data, or features—then rerun fixed validation cases.

Step 1

STEP 1 Simple tree: train 82%, validation 78%; gap 4 points.

Step 2

STEP 2 Deep tree: train 100%, validation 65%; gap 35 points.

Step 3

STEP 3 Deep tree's perfect **training** result is weaker unseen **evidence**.

Step 4

STEP 4 Prune, regularize, improve representative data, or features—then rerun fixed validation cases.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Overfitting and model capacity at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Constant	Training 5	Validation 6
Line	Training 2	Validation 3
Flexible curve	Training 0	Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Overfitting

See **overfitting** as a gap requiring investigation, not a label based only on model size.

The facts we will use

Supplied facts

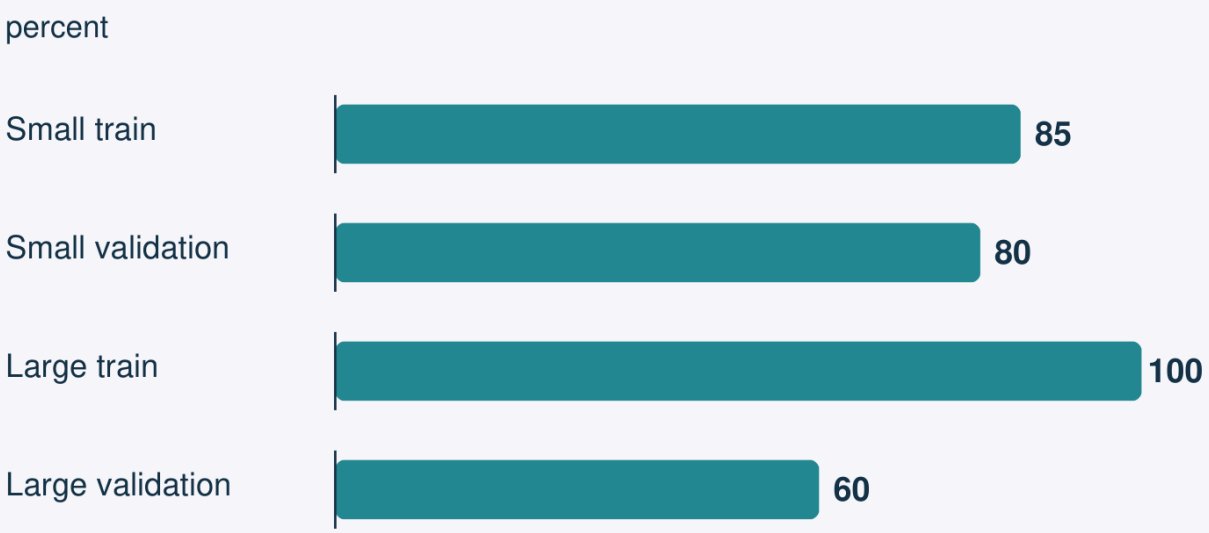
Toy model Small: **training** 85%, validation **80%**. Large: **training** 100%, validation **60%**. Same split and scoring rule.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/27

Training and validation accuracy



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/27

Overfitting

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute gaps

Small gap=5 percentage points; Large gap=40 percentage points.



2. Select development candidate

Small has the higher validation accuracy despite its lower training fit.



3. Investigate large model

Check capacity, training choices and data mismatch without tuning on final cases.

CHANGE ONE THING

Large's validation accuracy rises to82% after a development revision.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute gaps	Small gap=5 percentage points; Large gap=40 percentage points.
2. Select development candidate	Small has the higher validation accuracy despite its lower training fit.
3. Investigate large model	Check capacity, training choices and data mismatch without tuning on final cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Check **capacity**, **training** choices and data mismatch without tuning on final cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

A gap can also reflect data shift or sampling noise. Inspect **evidence** before attributing

every gap to **capacity**.

Keep the stated limit

Read the labelled picture: Does100% **training** accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Large's validation accuracy rises to82% after a development revision. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Overfitting claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The model with the highest training accuracy is the best model. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE The model with the highest **training** accuracy is the best model.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A simpler model is not always better. If a flexible candidate has better valid **evidence** under the declared rule, keep that **evidence** visible.

Model **capacity** is one part of an experiment, alongside features, sample size, label quality and the split.

What extra evidence would you inspect before concluding a model memorized noise?

STUDENT LESSON · 8 OF 18

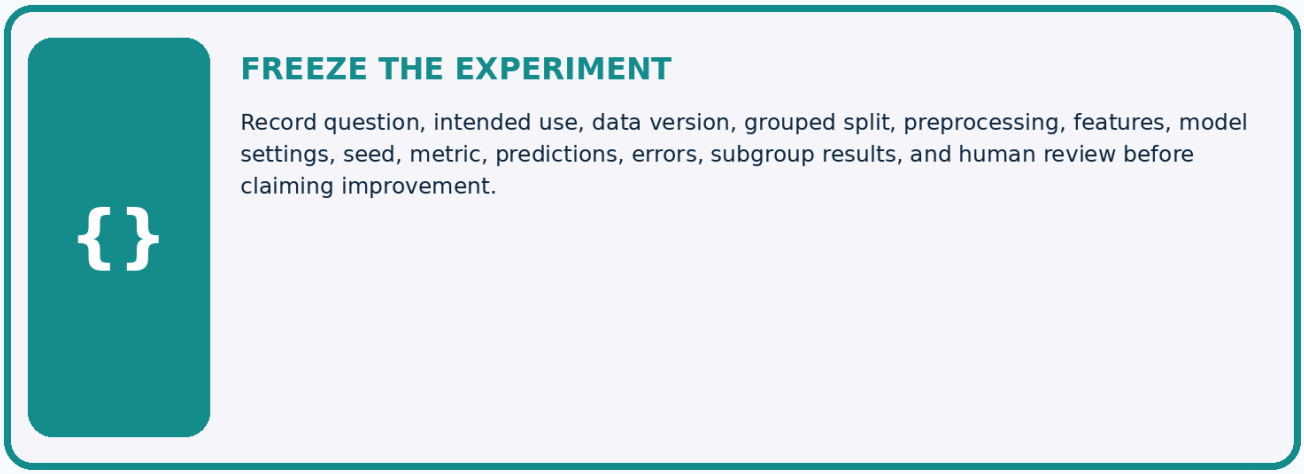
Practise together

Practise Overfitting with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Change one development factor at a time: simplify the candidate, improve data coverage or adjust a declared **regularization** setting. Refit on **training** and compare on validation.

Do not remove difficult valid examples or repeatedly search the test for a favorable result.

Practise with less help: Interpret a gap

Use workbook W2 for the connected example "**Overfitting** and model **capacity**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does100% **training** accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Overfitting. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Complexity Trial: fit paper trees of increasing depth, plot train/validation results, and identify the first clear overfit pattern.

1. Write the hypothesis and expected **evidence** before running.

2. **Complexity** Trial: fit paper trees of increasing depth, plot train/validation results, and identify the first clear overfit pattern.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

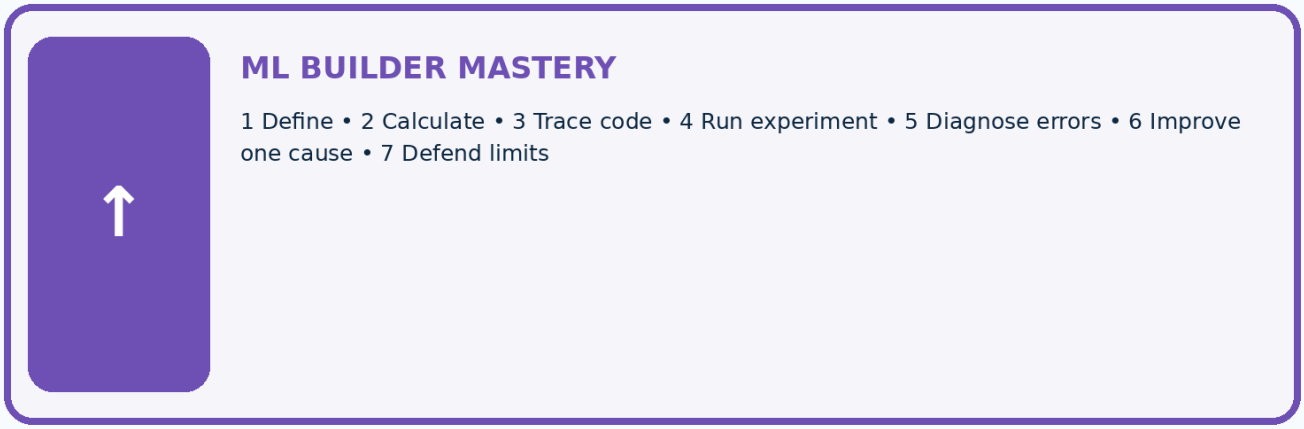
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Overfitting. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **overfitting**, **complexity**, **generalization gap**, **regularization**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE The model with the highest **training** accuracy is the best model.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

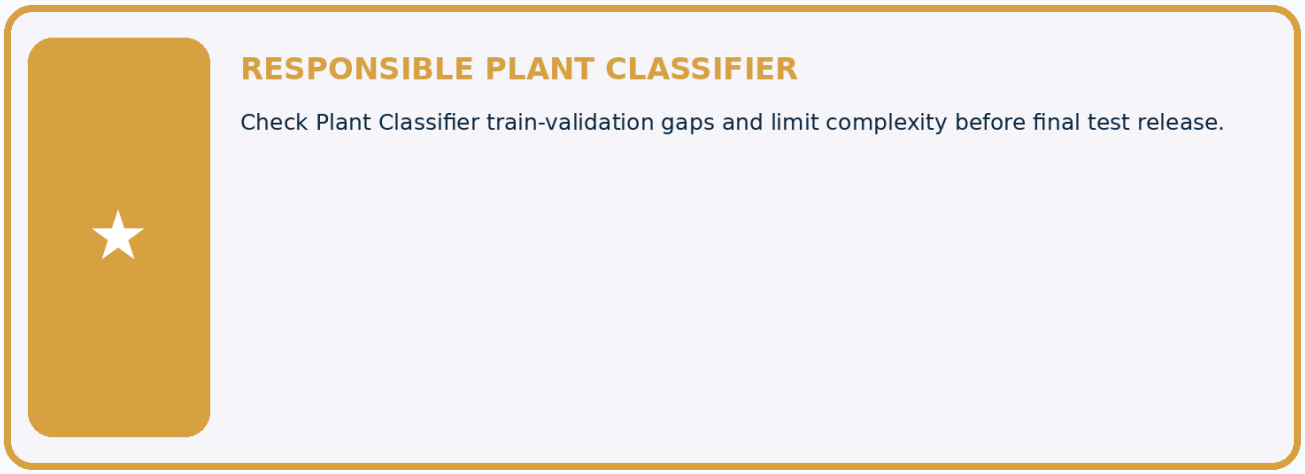
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Check Plant Classifier train-validation gaps and limit **complexity** before final test release.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

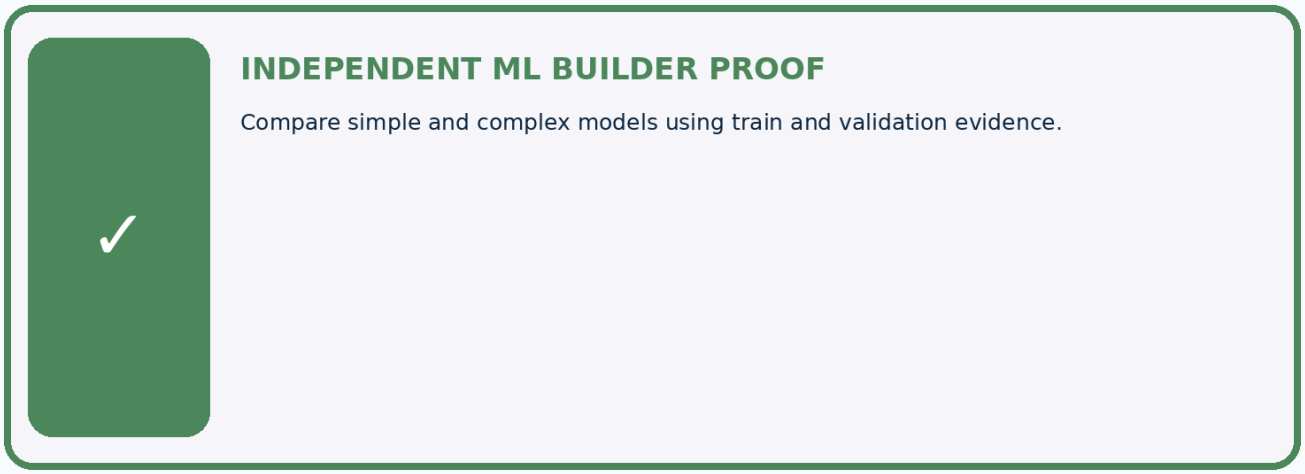
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Compare simple and complex models using train and validation **evidence**.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: Learning Curves.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Overfitting case for supported practice. Repeated examples do not become fresh assessments.

Week 27 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A scores100% train,**60%** validation; B scores85%,**80%** on the same sets.

The question

Compute gaps and select by validation accuracy.

Worked explanation

A gap40 percentage points; B gap5. B has better shared validation accuracy. A suggests **overfitting**, but investigate distribution differences too.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Compare training and shared validation

Evidence or stage	Value or result
A training	100
A validation	60
B training	85
B validation	80

A scores100% train,**60%** validation; B scores85%,**80%** on the same sets.

Compute gaps and select by validation accuracy.

A gap40 percentage points; B gap5. B has better shared validation accuracy. A suggests **overfitting**, but investigate distribution differences too.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

Model A **training**96%, validation62%; model B **training**88%, validation79%. Compare gaps and choose a development candidate, assuming the same valid evaluation conditions.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 27 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Model A has **training** accuracy 99% and validation accuracy **70%**; B has 88% and 84%. Compute each gap and discuss the **evidence**.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Check Plant Classifier train-validation gaps and limit **complexity** before final test release.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A has **training** 98%, validation 58%; B has **training** 86%, validation 81%. Compute gaps and select a candidate with a qualification.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Overfitting, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: See **overfitting** as a gap requiring investigation, not a label based only on model size.

By the end, I can	How I will show it
Compare training and validation evidence for model capacity	Explain the deciding step, show a trace or test, and state one limit.
Explain overfitting without assuming every gap proves it	Explain the deciding step, show a trace or test, and state one limit.
Use a declared selection rule instead of perfect training fit	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Overfitting without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Compare **training** and validation **evidence** for model **capacity**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

See **overfitting** as a gap requiring investigation, not a label based only on model size.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Check **capacity**, **training** choices and data mismatch without tuning on final cases.

1. Select development candidate

2. Compute gaps

3. Investigate large model

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Investigate large model Check **capacity**, **training** choices and data mismatch without tuning on final cases. Read the picture in order.

Check **capacity**, **training** choices and data mismatch without tuning on final cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Large's validation accuracy rises to82% after a development revision. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Its gap is 18 percentage points; it now beats Small on this validation set.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Its gap is 18 percentage points; it now beats Small on this validation set.

> Validation performance improved, but the **training**-validation gap is not a complete release rule.

Repair a missing building block

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train, Validation, Test (ai-4/24). Goal: Select a model using a declared validation rule. Short practice: Why does B's zero **training** MAE not settle the original choice?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as4.67 cm misleading?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

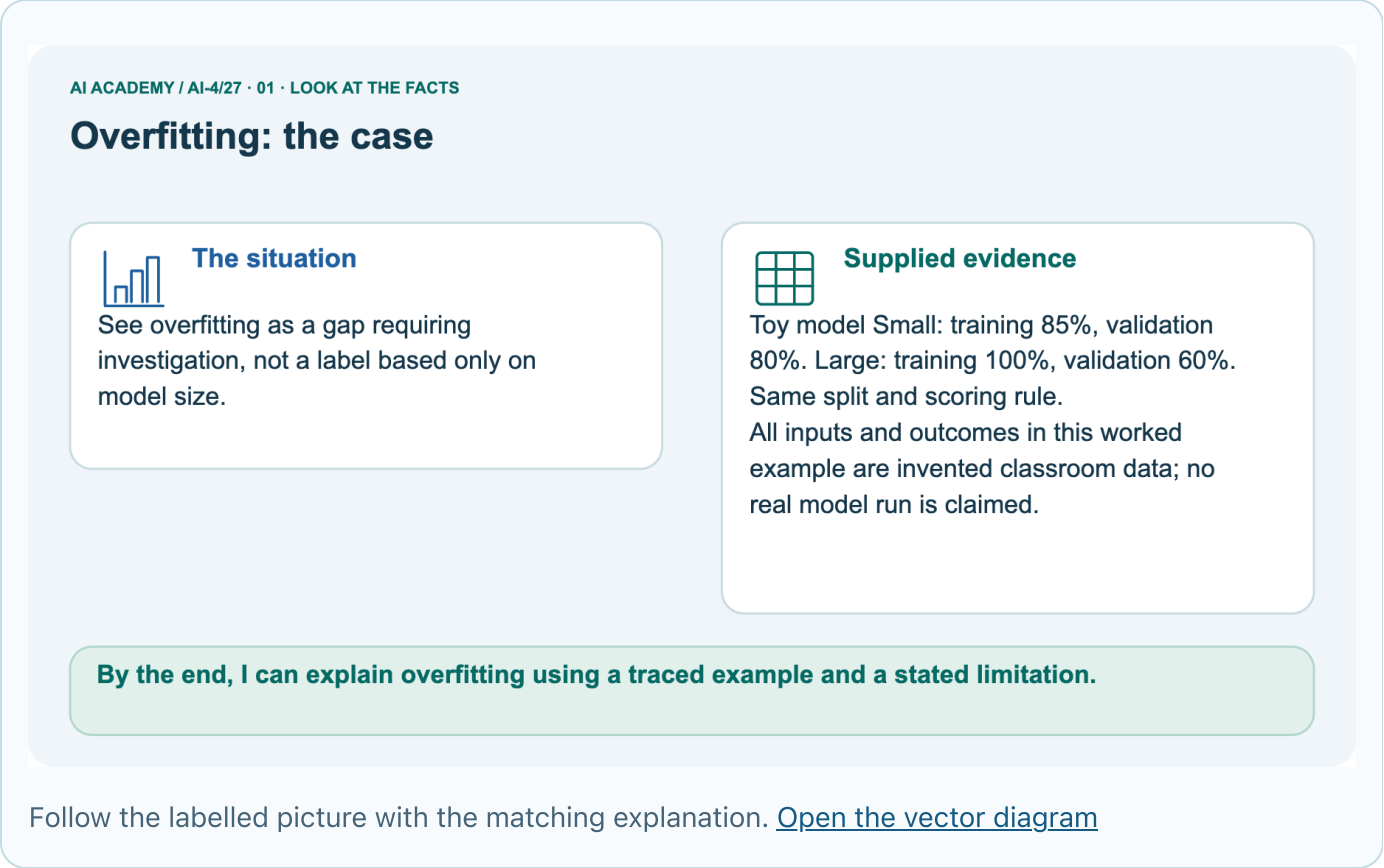
Drafts stay in this browser when storage is available; clear drafts on a shared device.

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Visual concept journey

> Explain each Overfitting graphic using its labels, deciding evidence and relevant limitation.

01. Overfitting: the case



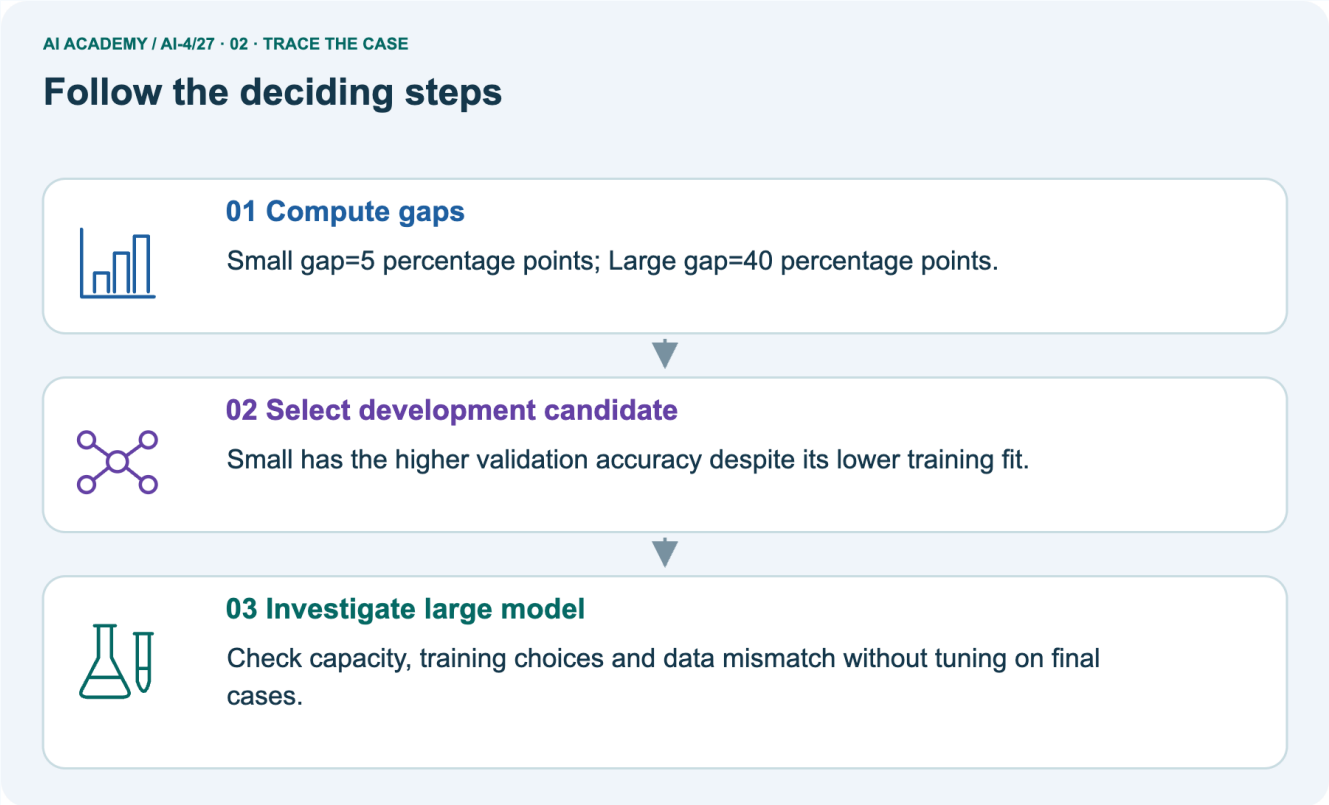
Follow the labelled picture with the matching explanation. Open the vector diagram

See **overfitting** as a gap requiring investigation, not a label based only on model size. Supplied facts: Toy model Small: **training** 85%, validation **80%**. Large: **training** 100%, validation **60%**. Same split and scoring rule.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

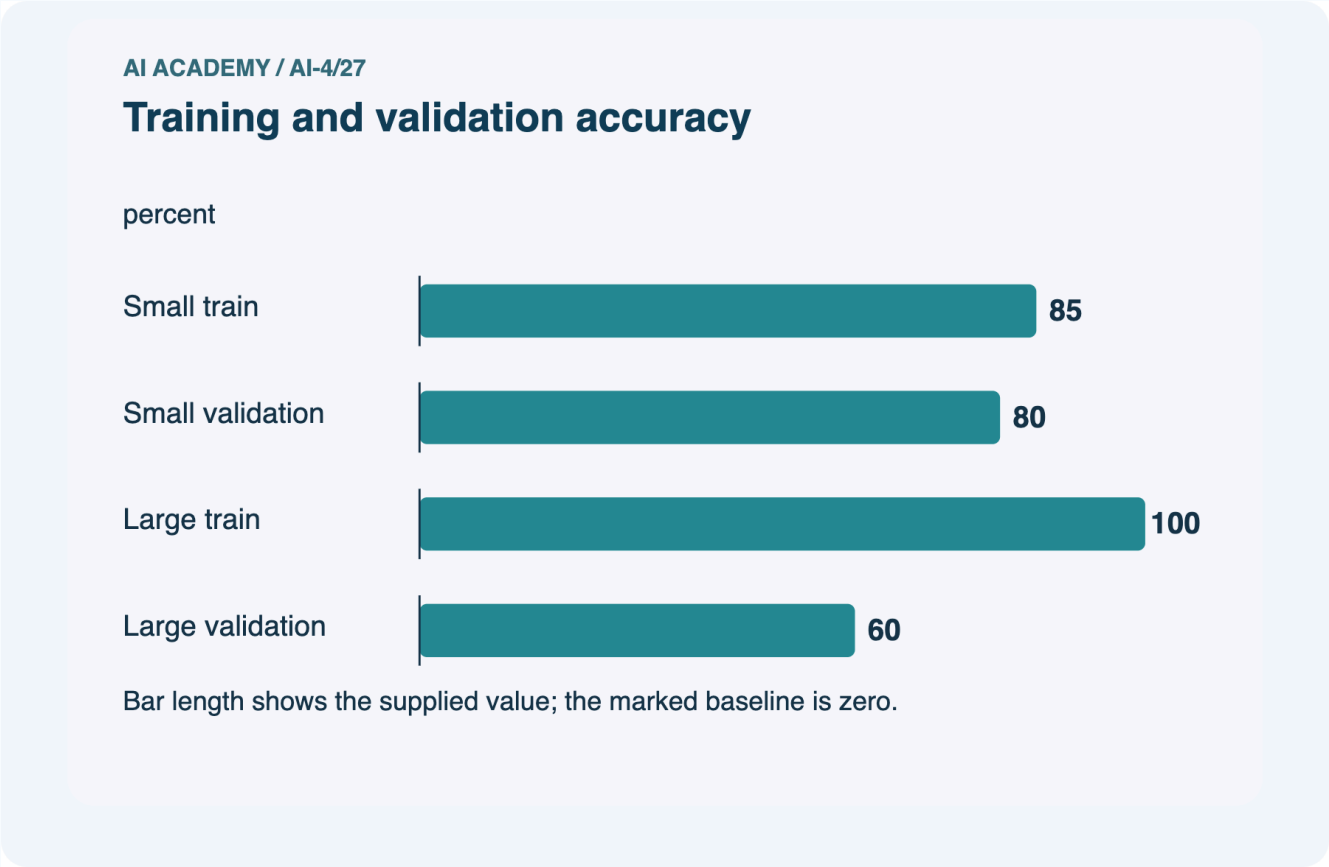


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute gaps: Small gap=5 percentage points; Large gap=40 percentage points. → Select development candidate: Small has the higher validation accuracy despite its lower **training** fit. → Investigate large model: Check **capacity**, **training** choices and data mismatch without tuning on final cases.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Training and validation accuracy

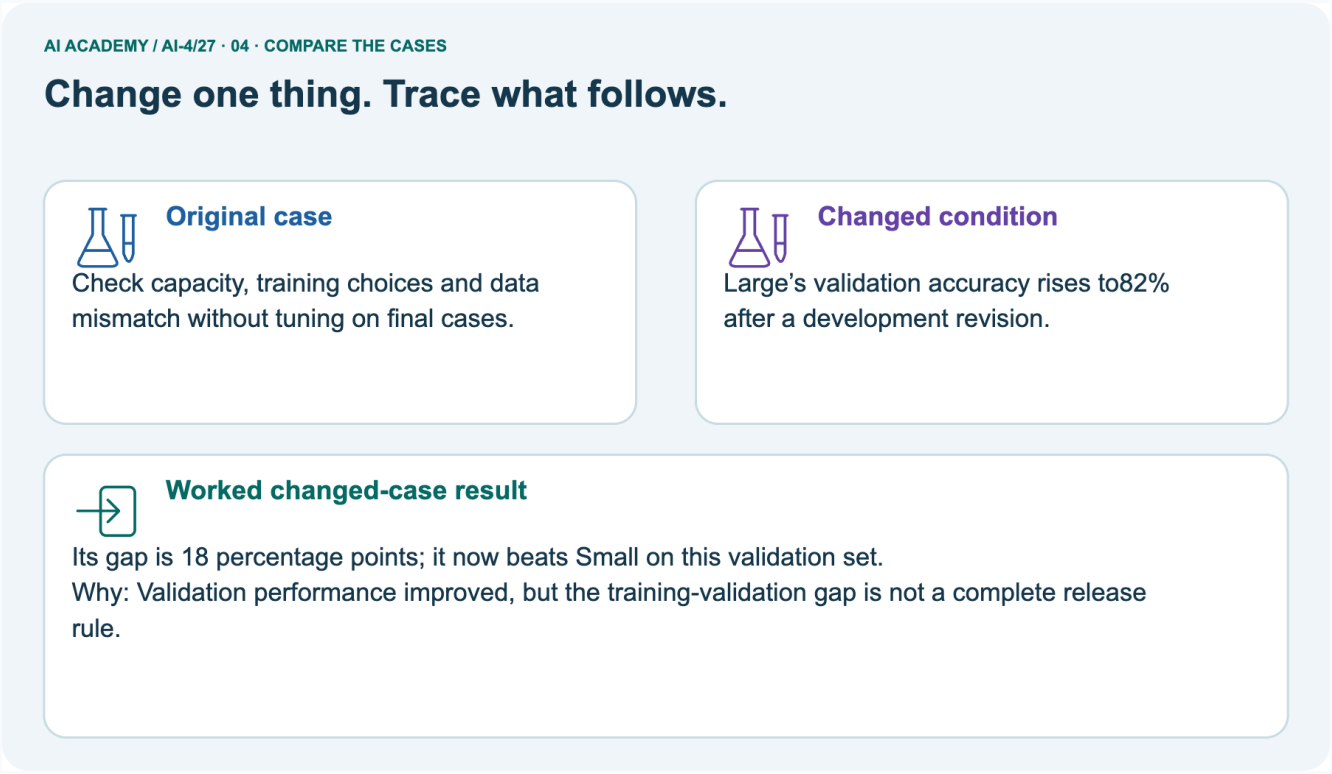


Follow the labelled picture with the matching explanation. Open the vector diagram

Training and validation accuracy. Units: percent. Small train: 85. Small validation: 80. Large train: 100. Large validation: 60.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



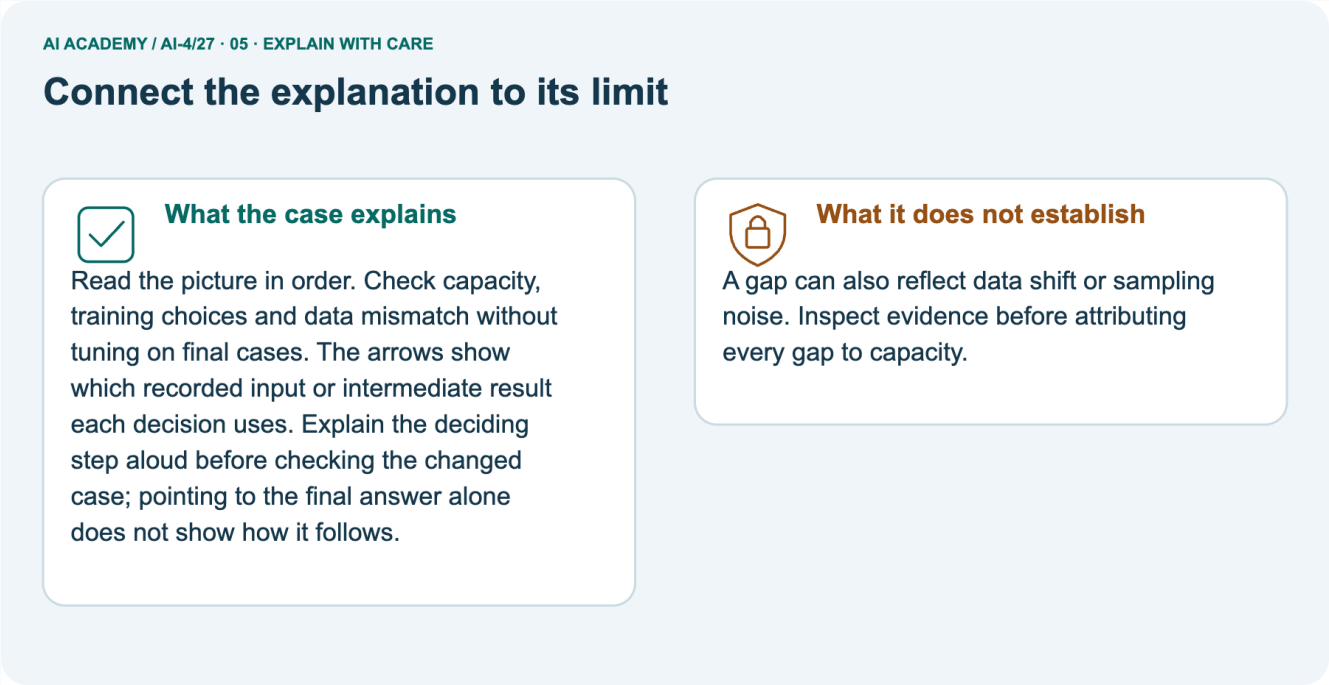
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Check **capacity**, **training** choices and data mismatch without tuning on final cases. Changed condition: Large's validation accuracy rises to82% after a development revision.

Expected result: Its gap is 18 percentage points; it now beats Small on this validation set. Reason: Validation performance improved, but the **training**-validation gap is not a complete release rule.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Check **capacity**, **training** choices and data mismatch without tuning on final cases. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: A gap can also reflect data shift or sampling noise.

Inspect **evidence** before attributing every gap to **capacity**.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does100% **training** accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.